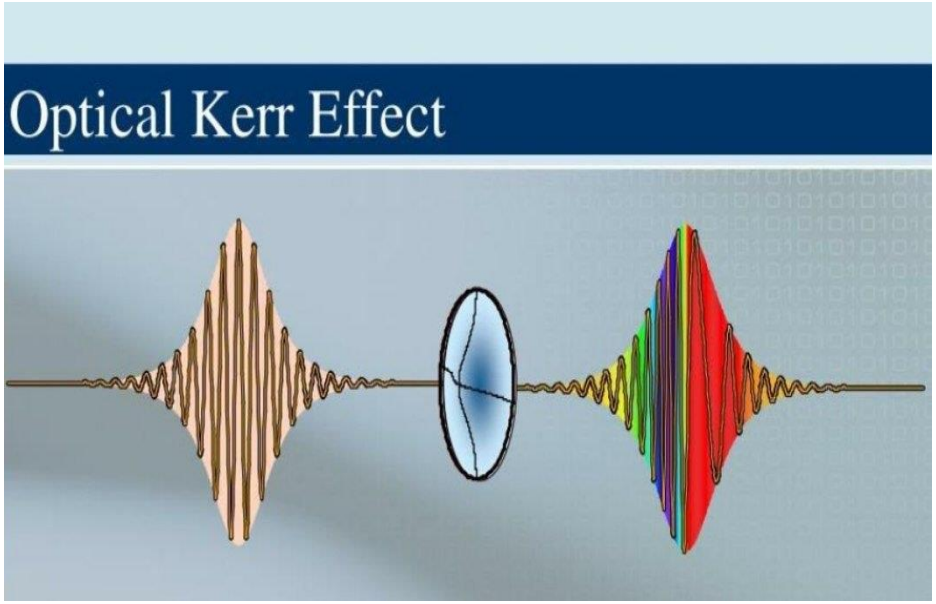
Nonlinear Optics

Lecture:4 & 5

Laser Engineering & Optoelectronics Department



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. Introduction to the Optical Kerr Effect

The Optical Kerr Effect :is a nonlinear optical phenomenon in which the refractive index of a material changes in response to the intensity of the incident light.

In linear optics, the refractive index is approximately independent of the intensity of the incident light. However, when intense laser light passes through a nonlinear medium, the refractive index can change.

This effect is commonly described by the following equation:

$$\boxed{n = n_0 + n_2 I}$$

Where:

- n: Refractive index under illumination.
- n_0: Linear refractive index.
- n_2: Nonlinear refractive index coefficient.
- I: Intensity of the incident light.

The equation shows that the refractive index depends on the intensity of the light.

Physical Principle of the Optical Kerr Effect

When an intense electromagnetic wave interacts with a material, the electric field can induce a nonlinear polarization.

The total polarization is expressed as:

$$P=\varepsilon 0[\chi(1)E+\chi(2)\boldsymbol{E^2}+\chi(3)\boldsymbol{E^3}+\cdots]$$

For the Optical Kerr Effect, the third-order polarization.

$$\mathbf{P^{(3)}=\varepsilon 0 X^{(3)} E^3}$$

$P^{(3)}$: Third-order nonlinear polarization.

$\varepsilon 0$: Permittivity of free space.

$\mathbf{X^{(3)}}$: Third-order nonlinear susceptibility.

E : Electric field of the incident light.

Types of Optical Kerr Effect

1. Intensity-Dependent Refractive Index

The refractive index changes with the intensity of the light passing through the material. This is described by

$$n = n_0 + n_2 \mathrm{I}$$

2. Kerr-Induced Birefringence

An applied optical field can produce different effective refractive indices for different polarization directions, resulting in an intensity-dependent phase difference.

Relationship Between Nonlinear Susceptibility and Refractive Index

The nonlinear refractive index coefficient is related to the real part of the third-order nonlinear susceptibility.
For a simple isotropic medium, using SI units and a specified =convention for the electric-field amplitude:n_2

$$n_2 = \frac{3\,\mathrm{Re}[\chi^{(3)}]}{4n_0^2\varepsilon_0 c}$$

Where:

- n_2: Nonlinear refractive index coefficient, in m^2/W
- $\mathrm{Re}[\chi^{(3)}]$: Real part of the third-order susceptibility.
- n_0: Linear refractive index.
- c: Speed of light in vacuum.
- ε_0: Permittivity of free space.

Problem 1: A material has a linear refractive index of 1.5 Its nonlinear refractive index coefficient is $2 \times 10^{-20}\ m^2\backslash W$,If the incident intensity is $5 \times 10^{18}\ W\backslash m^2$, **calculate the effective refractive index**.

Given: $n_0 = 1.5$, $n_2 = 2 \times 10^{-20}\ m^2\backslash W$, $I = 5 \times 10^{18}\ W\backslash m^2$

Equation:

$$n = n_0 + n_2 I$$

Solution:

$$n = 1.50 + (2 \times 10^{-20})(5 \times 10^{18})$$

$$n = 1.50 + 0.10$$

Final Answer:

$$\boxed{n = 1.60}$$

Applications of the Optical Kerr Effect

1.**Ultrafast Laser Technology**

The Kerr effect can be used in Kerr-lens mode locking to generate ultrashort laser pulses.

2. **Self-Focusing of Laser Beams**

When the nonlinear refractive index is positive, the intensity-dependent refractive index can cause a beam to focus itself under suitable conditions.

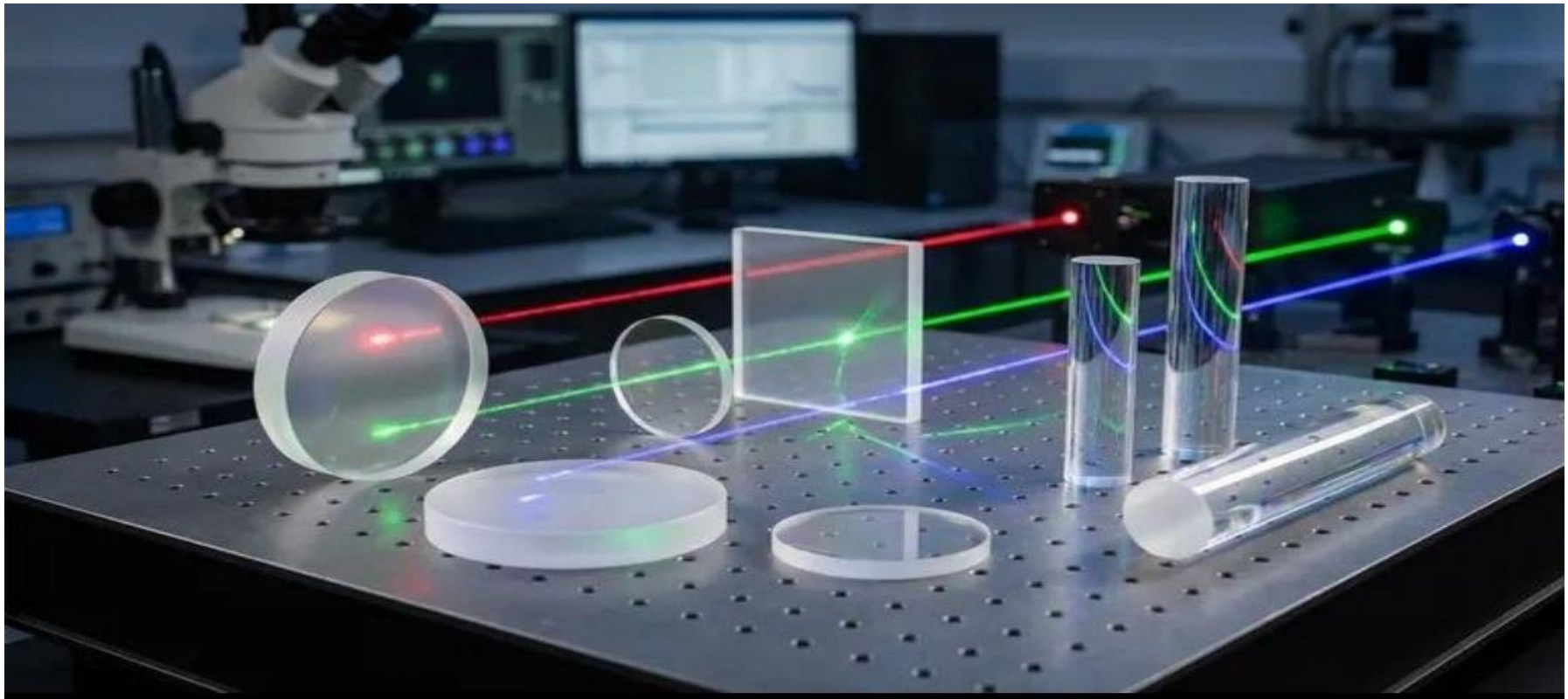
3. **Optical Communication**

The Kerr effect can be used in all-optical signal processing, but it can also cause nonlinear distortions in optical communication systems.

4. **Optical Kerr Shutters**

Kerr-induced birefringence can be used with polarizers to control the transmission of light on very short timescales.

Lecture: 5
THIRD-HARMONIC GENERATION (THG)



Introduction

Third-harmonic generation: is a nonlinear optical process in which light at an angular frequency ω interacts with a nonlinear medium and generates light at angular frequency **3ω**. In vacuum, the generated wavelength is one-third of the fundamental wavelength. THG is associated with the third-order nonlinear susceptibility $\chi^{(3)}$ and generally requires sufficiently intense optical fields.

Nonlinear Polarization

The polarization of a medium can be written as an expansion in powers of the electric field:

$$\mathbf{P = \varepsilon_0[\chi^{(1)}E + \chi^{(2)}E^2 + \chi^{(3)}E^3 + \ldots]}$$

Here, P is the induced polarization, ε_0 is the vacuum permittivity, E is the electric field, and $\chi^{(1)}$, $\chi^{(2)}$, and $\chi^{(3)}$ are the linear, second-order, and third-order susceptibilities.

Derivation of the Third-Harmonic Term

Let the incident field be

$\mathbf{E(t) = E_0 \cos(\omega t).}$

The third-order polarization is:

$\mathbf{P^{(3)}(t) = \varepsilon_0 \chi^{(3)} E_0^3 \cos^3(\omega t)}$

Using : $\mathbf{\cos^3(\omega t) = [3 \cos(\omega t) + \cos(3\omega t)]/4}$

gives:

$\mathbf{P^{(3)}(t) = (3/4)\varepsilon_0 \chi^{(3)} E_0^3 \cos(\omega t) + (1/4)\varepsilon_0 \chi^{(3)} E_0^3 \cos(3\omega t)}$

The term oscillating at $\mathbf{3\omega}$ drives the third-harmonic field. Thus, the generated nonlinear polarization component is proportional to $\chi^{(3)} E_0^3$.

Frequency and Wavelength

The third-harmonic frequency is

$\mathbf{f_3 = 3f}$.

Since $c = \lambda f$ in vacuum,

the third-harmonic wavelength is

$\boldsymbol{\lambda_3 = \lambda/3}$.

Example : A 1500 nm laser undergoes THG. Find the generated wavelength.

Solution: $\lambda_3 = 1500 \text{ nm} / 3 = 500 \text{ nm}$.

Phase Matching

Efficient conversion requires the fundamental and generated waves to maintain a suitable phase relationship as they propagate.

The phase mismatch is $\Delta k = k_{3\omega} - 3k_{\omega}$.

Ideal phase matching occurs when **$\Delta k = 0$**.

If phase matching is **not satisfied**, contributions generated at different positions may interfere destructively and limit the conversion efficiency.

Applications

1.Nonlinear microscopy for imaging interfaces and biological or transparent materials.

2.Characterization of nonlinear optical properties of materials.

3.Generation of shorter-wavelength radiation from laser sources.

4.Ultrafast optics and studies of intense laser pulses.

5.Optical diagnostics of interfaces and inhomogeneous media.

Review Questions

1.What is third-harmonic generation?

2.Why is THG a third-order nonlinear process?

3.A laser wavelength is 900 nm. Calculate the third-harmonic wavelength.

4.Define the phase mismatch Δk.

5. Mention three applications of THG.